

Project Synopsis: Scaling ESAP for Modern High-Density Sensor Data

Background

We are working with an implementation of the **ESAP (Electrical Conductivity Sampling Assessment Program) / Spatial Response Surface (SRS)** methodology described by Lesch (2005), which is widely used for selecting optimal soil sampling locations based on sensor data (e.g., EM38).

The core idea of ESAP/SRS is to:

1. Transform sensor data using **PCA** to reduce dimensionality.
 2. Define an idealized **response surface design** in PCA space.
 3. Select field sampling locations that:
 - Match these PCA targets.
 - Are **maximally spatially distributed** to ensure statistical independence.
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Problem: ESAP Does Not Scale to Modern Data

The traditional ESAP implementation relies heavily on **full pairwise distance calculations**:

- Uses operations like:
 - `distance_matrix()`
 - `cdist()`
- Computes distances between **all points and all other points**.

! Computational Implications:

- **Memory complexity:** $O(n^2)$
- **Time complexity:** $O(n^2)$

This becomes prohibitive for modern datasets:

- Drone imagery (10,000–1,000,000+ pixels)
- High-resolution proximal sensors
- Dense EM38 surveys

Result:

- Crashes due to RAM limits
 - Extremely slow runtimes
 - Not feasible for real-world high-resolution applications
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Why This Matters

Modern agricultural and environmental monitoring increasingly rely on:

- **Drone-based multispectral imagery (NDVI, NDRE, etc.)**
- **High-density proximal sensing (EM38, Veris, etc.)**
- **Edge-of-field and watershed-scale monitoring**

These datasets are:

- Dense
- Spatially structured
- Information-rich

➡ But **current ESAP implementations cannot handle them efficiently**, limiting their practical use.

💡 Proposed Solution: Nearest-Neighbor (NN) Approximation

We are replacing full pairwise distance computations with **nearest-neighbor methods** using structures like:

- **KDTree**
- **BallTree**

☑ Benefits:

- **Memory:** $O(n \log n)$ instead of $O(n^2)$
- **Speed:** Dramatically faster queries
- **Scalable:** Works with 100k–1M+ points

⚖️ Key Question

Does replacing full pairwise distance optimization with **nearest-neighbor-based spacing**:

- ? Change the methodology?
- ? Or simply improve performance?

🧠 Working Hypothesis

Nearest-neighbor methods can approximate ESAP's spatial optimization **without loss of validity**, under reasonable assumptions:

Assumptions:

1. **Data are uniformly or systematically spaced**
 - Drone pixels (gridded)
 - EM38 transects (structured)
2. **Spatial correlation is short-range**
 - Soil and vegetation properties often decorrelate within ~30–60 m
3. **Residuals are locally structured**

- PCA + regression captures global trends

4. PCA space is well-behaved

- Continuous, convex, and sufficiently sampled

➡ Under these conditions:

- **Local spacing constraints $\approx$ global spacing optimization**
- Nearest-neighbor distance becomes a proxy for full pairwise separation

Critical Challenge: Choosing Optimal NN Distance

Instead of arbitrarily choosing a spacing threshold (e.g., 30–60 m), we want to **estimate the optimal nearest-neighbor distance (d_{nn})**.

Candidate Methods:

1. Empirical Optimization

- Sweep NN distances (20–100 m)
- Evaluate:
 - Model performance (R^2 , RMSE)
 - Residual spatial autocorrelation (Moran's I)
- Select threshold where:
 - Moran's I ≈ 0
 - Performance stabilizes

2. Variogram-Based Approach

- Fit variogram to residuals
- Use ~95% of spatial range as NN distance

3. Heuristic (Low-Memory)

- Analyze local variance across k-nearest neighbors
- Choose distance where variance stabilizes

4. Hybrid Validation

- Compare NN-based sampling vs. full ESAP on subset:
 - geoMSD (spatial spread)
 - Moran's I (independence)
 - Prediction accuracy

Goal of Next Steps

We want to:

1. Implement **NN-based sampling** in place of full pairwise distance methods.
 2. Determine the **optimal NN distance** using one or more strategies above.
 3. Validate that:
 - Spatial independence is preserved
 - Model performance is unchanged
 4. Achieve:
 - **Massive reductions in RAM usage**
 - **Scalable performance for large datasets**
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End Goal

Create a **modernized ESAP workflow** that:

- Maintains scientific validity
 - Works efficiently on large datasets (drone, EM38, etc.)
 - Enables practical field deployment at scale
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What I Need Help With Next

- Implementing NN-based sampling logic
- Building functions to estimate optimal NN distance
- Validating NN vs. full pairwise methods
- Ensuring statistical assumptions remain satisfied